# JAVASCRIPT ASSIGNMENT : 02

## Module 1 JavaScript Introduction Assignment Questions

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**1. Explain JavaScript in your own words with real world example**

**Write explanation of JavaScript and explain how it makes websites interactive with at least three practical examples.**
**Hint: Think about form validation, button click events and dynamic content updates.**

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**2. Explain history of JavaScript and its evolution**

**Write short note on how JavaScript was created, who created it and how it evolved over time. Explain why it became popular.**
**Hint: Mention browser competition and standardization.**

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**3. Compare JavaScript and Java with practical differences**

**Create comparison table explaining differences between JavaScript and Java in terms of language type, execution, use case and syntax.**
**Hint: Focus on compiled vs interpreted and browser vs application usage.**

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**4. Explain role of JavaScript in modern web development**

**Explain how JavaScript is used in frontend and backend today. Include examples like dynamic UI and API communication.**
**Hint: Mention interaction with HTML and CSS.**

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**5. Explain client side JavaScript with examples**

**Explain what client side JavaScript means and give practical examples where it is used.**
**Hint: Think about browser execution and DOM manipulation.**

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**6. Explain server side JavaScript with example**

**Explain how JavaScript can run on server and what tasks it performs.**
**Hint: Mention runtime like Node and backend operations.**

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**7. Compare client side and server side JavaScript**

**Write comparison explaining execution environment, purpose and advantages.**
**Hint: Think browser vs server difference.**

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**8. Explain JavaScript engine and how it works**

**Explain what JavaScript engine is and how it executes code internally.**
**Hint: Mention parsing, compiling and execution.**

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**9. Explain V8 engine in simple terms**

**Explain what V8 engine is and why it is important in JavaScript ecosystem.**
**Hint: Mention Chrome browser and performance.**

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**10. Write short note on why JavaScript is called scripting language**

**Explain why JavaScript is considered scripting language and how it differs from traditional programming languages.**
**Hint: Think about runtime environment and embedding.**

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**11. Explain why JavaScript is single threaded**

**Explain concept of single thread and how JavaScript handles multiple tasks.**
**Hint: Mention event loop concept briefly.**

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**12. Explain how JavaScript interacts with HTML and CSS**

**Explain how JavaScript controls webpage structure and style dynamically.**
**Hint: Think about DOM and style manipulation.**

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**13. Write advantages and limitations of JavaScript**

**Write at least five advantages and three limitations of JavaScript.**
**Hint: Think about browser compatibility and security.**

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**14. Explain how JavaScript became full stack language**

**Explain how JavaScript moved from browser only language to full stack development.**
**Hint: Mention backend runtime and frameworks.**

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**15. Create conceptual diagram explanation of JavaScript ecosystem**

**Draw or explain structure of JavaScript ecosystem including browser, engine, frontend frameworks and backend runtime.**
**Hint: Structure explanation clearly step by step.**

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